ER Diagram - Easy Revision Guide

A short but deep explanation of Entity-Relationship diagrams with one complete college example.

ER Diagram = Visual blueprint of a database.
It shows what data we store, the properties of that data, and how different parts of the database are connected.

1. The 3 Core Parts

Part	Meaning	Example
Entity	A real-world thing/concept about which data is stored. It normally becomes a table.	Student, Department
Attribute	A property that describes an entity. It normally becomes a column.	StudentID, Name, Email
Relationship	Shows how two entities are connected.	Department has Students

2. Complete Example: College Database

DEPARTMENT
DepartmentID (PK)
DepartmentName

1 ---- has ---- M

STUDENT
StudentID (PK)
Name
Email
DepartmentID (FK)

How to read it: One Department can have many Students. DepartmentID is the Primary Key in Department and appears as a Foreign Key in Student to create the connection.

3. Cardinality - How Many?

Type	Meaning	Easy Example	Database idea
1 : 1	One to One	Person - Passport	Foreign Key + UNIQUE
1 : M	One to Many	Department - Students	Foreign Key on the many side
M : N	Many to Many	Students - Courses	Use a junction/bridge table

4. Many-to-Many: Why a Bridge Table?

A student can study many courses, and a course can contain many students. We do not store a list of CourseIDs inside one Student cell. Instead, we create a third table such as **Enrollment**.

STUDENT
StudentID (PK)
Name

1 : M

ENROLLMENT
StudentID (FK)
CourseID (FK)

M : 1

COURSE
CourseID (PK)
CourseName

5. Primary Key vs Foreign Key in an ER Diagram

Primary Key (PK)	Uniquely identifies each row inside its own entity/table. Example: StudentID.
Foreign Key (FK)	Stores a key from another table so the two entities can be connected. Example: Student.DepartmentID -> Department.DepartmentID.

6. How to Build an ER Diagram - 5 Easy Steps

1. Identify the main things you need to store - Student, Course, Department.
2. List their attributes - ID, Name, Email, etc.
3. Choose a Primary Key for every entity.
4. Decide relationships and cardinality - 1:1, 1:M, or M:N.
5. Add Foreign Keys or a bridge table so the design can become real SQL tables.

7. Common Mistakes

Mistake	Better design
Using Name as the main relationship key	Use a stable ID such as StudentID or DepartmentID.
Putting 101,102,103 in one CourseID cell	Use a bridge table for many-to-many relationships.
Creating tables before thinking about relationships	Design the ER structure first, then write CREATE TABLE.

Fast Memory: Entity = Table | Attribute = Column | Relationship = Connection | Cardinality = How many | PK = Unique identity | FK = Connection to another table | M:N = Bridge table